

ELEN 635
Review for Final Exam

The exam is closed book and closed notes and no calculator. You will be provided with the attached equation sheet. I will provide paper.

Distance Ed Instructions: All distance learning students with a local address must take the exam with the in class students. The exam will be held at a time convenient to the greatest number of students. I will email you shortly to give you a suggested time. The exam will be posted on Canvas, and open 8 minutes early. You then print out the exam. After two hours the test will shut off on Canvas and you will have 10 minutes take it to the printer and upload it to the Canvas Test module. The procedure is the same as it is with homework. Failure to upload in 10 minutes will incur a serious points penalty. Your cell phone should be on at the website used for office hours the entire time you are taking the exam. It should be propped up horizontally so that you and the screen are visible. Take the cell phone with you to the printer. I will come on to the Zoom site early so you can set up.

The following is a list of topics which cover the types of problems that will be asked on the final exam. The final will cover all material since the first exam – Starting with the Reciprocity Theorem in Notes 7 and material since then.

1. Starting with Maxwell's equations, derive the reciprocity equation:
$$\oint_s (\mathbf{E}^b \times \mathbf{H}^a - \mathbf{E}^a \times \mathbf{H}^b) \cdot d\mathbf{s} = \int_{vol} (\mathbf{E}^a \cdot \mathbf{J}^b - \mathbf{E}^b \cdot \mathbf{J}^a + \mathbf{H}^b \cdot \mathbf{M}^a - \mathbf{H}^a \cdot \mathbf{M}^b) dv$$
2. Given the general form of the reciprocity theorem prove that an impressed electric current an infinitesimal distance above the surface of a perfect electric conductor radiates no field.
3. Prove that the field in any given volume is unique.
4. Use the equivalence principle to obtain the fields radiated by a PEC dipole antenna.
5. Use physical optics to obtain the fields scattered by a small PEC or PMC plate.
6. Given $\nabla'^2 A + k^2 A = -J(\mathbf{r}')$ and Green's identities, derive $A(\mathbf{r}) = \int_{v'} J(\mathbf{r}') G(\mathbf{r}, \mathbf{r}') dv'$.
7. Find the mode functions ψ and the expressions for the fields in terms of the mode functions for TE or TM cases in a rectangular waveguide with or without a mixed dielectric/air region.
8. Find the fields in a waveguide with an electric or magnetic current source.
9. Obtain the mode functions and the expressions for the fields in terms of the mode functions for TE or TM cases in a grounded slab waveguide.
10. Find the equations necessary to obtain the TE or TM modes in a grounded dielectric slab waveguide.
11. Microstrip Antenna

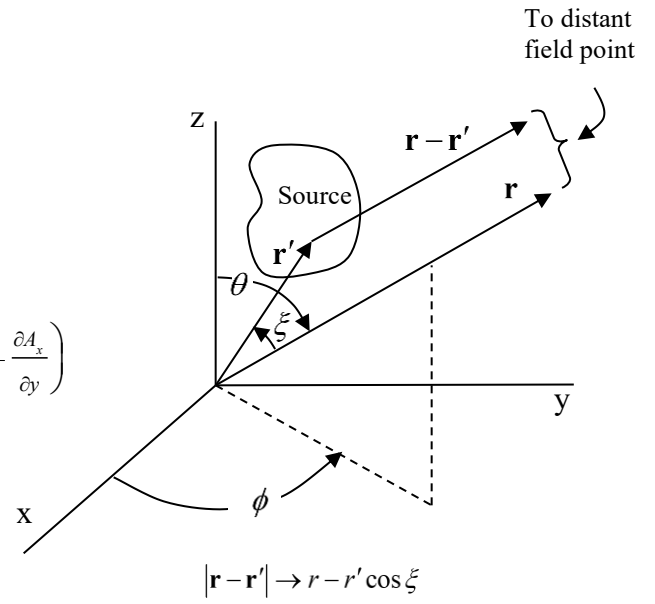
Cartesian Coordinates (x, y, z)

$$\nabla V = \mathbf{a}_x \frac{\partial V}{\partial x} + \mathbf{a}_y \frac{\partial V}{\partial y} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \mathbf{a}_x \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \mathbf{a}_y \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \mathbf{a}_z \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$



Cylindrical Coordinates (r, phi, z)

$$\nabla V = \mathbf{a}_r \frac{\partial V}{\partial r} + \mathbf{a}_\phi \frac{\partial V}{r \partial \phi} + \mathbf{a}_z \frac{\partial V}{\partial z}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{\partial A_\phi}{r \partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \mathbf{A} = \mathbf{a}_r \left(\frac{\partial A_z}{r \partial \phi} - \frac{\partial A_\phi}{\partial z} \right) + \mathbf{a}_\phi \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) + \mathbf{a}_z \left[\frac{\partial}{\partial r} (r A_\phi) - \frac{\partial A_r}{\partial \phi} \right]$$

$$\nabla^2 V = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

$$r' \cos \xi = (x' \cos \phi + y' \sin \phi) \sin \theta + z' \cos \theta \quad (3-100)$$

$$r' \cos \xi = \rho' \sin \theta \cos(\phi - \phi') + z' \cos \theta \quad (3-101)$$

$$r' \cos \xi = r' [\cos \theta \cos \theta' + \sin \theta \sin \theta' \cos(\phi - \phi')] \quad (3-102)$$

$$A^2 = \mathbf{A} \cdot \mathbf{A}$$

$$|A|^2 = \mathbf{A} \cdot \mathbf{A}^*$$

$$\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$$

$$\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$$

$$\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$$

$$(\mathbf{A} + \mathbf{B}) \cdot \mathbf{C} = \mathbf{A} \cdot \mathbf{C} + \mathbf{B} \cdot \mathbf{C}$$

$$\mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \mathbf{B} \cdot \mathbf{C} \times \mathbf{A} = \mathbf{C} \cdot \mathbf{A} \times \mathbf{B}$$

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C}) \mathbf{B} - (\mathbf{A} \cdot \mathbf{B}) \mathbf{C}$$

$$\nabla(v + w) = \nabla v + \nabla w$$

$$\nabla \cdot (\mathbf{A} + \mathbf{B}) = \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B}$$

$$\nabla \times (\mathbf{A} + \mathbf{B}) = \nabla \times \mathbf{A} + \nabla \times \mathbf{B}$$

$$\nabla(vw) = w \nabla v + v \nabla w$$

$$\nabla \cdot (w \mathbf{A}) = w \nabla \cdot \mathbf{A} + \mathbf{A} \cdot \nabla w$$

$$\nabla \times (w \mathbf{A}) = w \nabla \times \mathbf{A} - \mathbf{A} \times \nabla w$$

$$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot \nabla \times \mathbf{A} - \mathbf{A} \cdot \nabla \times \mathbf{B}$$

$$\nabla^2 \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla \times \nabla \times \mathbf{A}$$

$$\nabla \times (v \nabla w) = \nabla v \times \nabla w$$

$$\nabla \times \nabla w = 0$$

$$\nabla \cdot \nabla \times \mathbf{A} = 0$$

Spherical Coordinates (R, theta, phi)

$$\nabla V = \mathbf{a}_R \frac{\partial V}{\partial R} + \mathbf{a}_\theta \frac{\partial V}{R \partial \theta} + \mathbf{a}_\phi \frac{1}{R \sin \theta} \frac{\partial V}{\partial \phi}$$

$$\nabla \cdot \mathbf{A} = \frac{1}{R^2} \frac{\partial}{\partial R} (R^2 A_R) + \frac{1}{R \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{R \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\nabla \times \mathbf{A} = \mathbf{a}_R \frac{1}{R \sin \theta} \left[\frac{\partial}{\partial \theta} (A_\phi \sin \theta) - \frac{\partial A_\theta}{\partial \phi} \right]$$

$$+ \mathbf{a}_\theta \frac{1}{R} \left[\frac{1}{\sin \theta} \frac{\partial A_R}{\partial \phi} - \frac{\partial}{\partial R} (R A_\phi) \right]$$

$$+ \mathbf{a}_\phi \frac{1}{R} \left[\frac{\partial}{\partial R} (R A_\theta) - \frac{\partial A_R}{\partial \theta} \right]$$

$$\nabla^2 V = \frac{1}{R^2} \frac{\partial}{\partial R} \left(R^2 \frac{\partial V}{\partial R} \right) + \frac{1}{R^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{R^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$

$$\iiint \nabla \cdot \mathbf{A} dv = \oint \mathbf{A} \cdot d\mathbf{s}$$

$$\iint \nabla \times \mathbf{A} \cdot d\mathbf{s} = \oint \mathbf{A} \cdot d\mathbf{l}$$

$$\nabla^2\mathbf{A}+k^2\mathbf{A}=\mathbf{0}$$

$$\mathbf{E}=\frac{-j\omega\mu}{k^2}\big(k^2\mathbf{A}+\nabla\nabla\cdot\mathbf{A}\big)$$

$$\mathbf{H}=\nabla\times\mathbf{A}$$

$$\nabla^2\mathbf{F}+k^2\mathbf{F}=\mathbf{0}$$

$$\mathbf{E}=-\nabla\times\mathbf{F}$$

$$\mathbf{H}=\frac{-j\omega\mathcal{E}}{k^2}\big(k^2\mathbf{F}+\nabla\nabla\cdot\mathbf{F}\big)$$

$$\int\limits_V\Big(\nabla\psi\cdot\nabla\phi+\psi\nabla^2\phi\Big)dv=\oint\limits_s\psi\,\frac{\partial\phi}{\partial n}\,ds$$

$$\int\limits_V\Big(\psi\nabla^2\phi-\phi\nabla^2\psi\Big)dv=\oint\limits_S\Bigg(\psi\,\frac{\partial\phi}{\partial n}-\phi\,\frac{\partial\psi}{\partial n}\Bigg)ds$$